Ancient Clifford-Product Mean-Curvature Flow: Exact Lifespans, Focal Cylinders, and Minimal Stability

Route-A source-local certificate HCS-C319

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Abstract

For every $p, q \geq 1$, we solve mean-curvature flow inside the unit sphere for the full parallel family $S^p(\cos\theta) \times S^q(\sin\theta)$. The affine coordinate $y = \sin^2\theta$ gives both ancient branches, their exact finite lifespans, and their focal collapses.

1 Exact phase portrait and lifespan

Put $n = p + q$, and write

$$\Sigma_\theta = S^p(\cos\theta) \times S^q(\sin\theta) \subset S^{n+1}, \qquad 0 < \theta < \frac{\pi}{2}.$$

For $x \in S^p$, $z \in S^q$, choose $\nu = (-\sin\theta\, x, \cos\theta\, z)$. Our scalar convention is $\partial_t F = H\nu$; this is the negative area gradient for the signs below.

Theorem 1 (Complete product-family flow)**.** *The principal curvatures are* $\tan\theta$ *with multiplicity* p *and* $-\cot\theta$ *with multiplicity* q*. Hence*

$$H = p\tan\theta - q\cot\theta, \qquad y' = 2(ny - q), \qquad y = \sin^2\theta.$$

The unique stationary leaf has $y_* = q/n$. Every other solution is ancient,

$$y(t) = \frac{q}{n} + \left(y_0 - \frac{q}{n}\right) e^{2nt}, \tag{1}$$

and has one finite forward endpoint. If $y_0 < q/n$, then

$$T_- = \frac{1}{2n} \log \frac{q}{q - ny_0}, \qquad y(t) \downarrow 0,$$

and the S^p factor survives as the focal submanifold. If $y_0 > q/n$, then

$$T_+ = \frac{1}{2n} \log \frac{p}{ny_0 - q}, \qquad y(t) \uparrow 1,$$

and the S^q factor survives. In both cases $y(t) \to q/n$ as $t \to -\infty$.

Proof. Differentiating $F = (\cos\theta\, x, \sin\theta\, z)$ gives $\partial_\theta F = \nu$. The two shape eigenvalues follow by differentiating ν along each factor. Thus $\theta' = H$, and multiplication by $2\sin\theta\cos\theta$ yields the affine equation for y. Solving it gives (1); imposing $y = 0$ or $y = 1$ gives the stated times. Their logarithm arguments exceed one on the corresponding branches. Backward convergence, uniqueness of the stationary leaf, and the focal labels now follow directly. □

The faces $y = 0, 1$ are focal submanifolds, not regular product hypersurfaces. The cases $p = 0$ or $q = 0$ are excluded degenerate sphere/double-cover geometries. The backward endpoint is at infinite time, not an added slice.

Source lineage

References

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[2] H.-L. Liu and C.-L. Terng, "The mean curvature flow for isoparametric submanifolds," *Duke Mathematical Journal* 147 (2009), 157–179. DOI: 10.1215/00127094-2009-009.